

Here's a clean and effective Multi-Area OSPF practice lab brief:

Multi-Area OSPF Practice Lab – GlobalTech Corporation

Objective:

Configure **Multi-Area OSPF** to provide efficient routing across a growing company with multiple locations.

Topology Overview:

- **Area 0** (Backbone) – Core network
- **Area 1** – Headquarters
- **Area 2** – Branch Office A
- **Area 3** – Branch Office B

Devices:

- 4 Routers (R1, R2, R3, R4)
- 1 ISP Router
- 4 PCs (one per LAN)

IP Addressing & Area Design:

Router	Interface	IP Address	Area	Purpose
R1	G0/0	172.16.10.1/24	1	HQ LAN
R1	G0/1	172.16.1.1/30	0	Link to R2
R1	G0/2	203.0.113.1/30	—	To ISP
R2	G0/0	172.16.20.1/24	2	Branch A LAN
R2	G0/1	172.16.1.2/30	0	Link to R1
R2	G0/2	172.16.2.1/30	0	Link to R3
R3	G0/0	172.16.30.1/24	3	Branch B LAN

R3	G0/1	172.16.2.2/30	0	Link to R2
R4	G0/0	172.16.40.1/24	1	Another HQ segment

PCs:

- PC1: 172.16.10.10 (VLAN 10, DG: 172.16.10.1)
 - PC2: 172.16.20.10 (VLAN 20, DG: 172.16.20.1)
 - PC3: 172.16.30.10 (VLAN 30, DG: 172.16.30.1)
 - PC4: 172.16.40.10 (VLAN 40, DG: 172.16.40.1)
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Tasks:

1. Configure all interfaces with correct IP addresses.
 2. Configure **Multi-Area OSPF**:
 - Area 0 as the backbone
 - Area 1, 2, and 3 as normal areas
 3. Make R1 the **ASBR** by configuring a default route to the ISP and propagating it into OSPF.
 4. Use proper **passive-interface** on all LAN interfaces.
 5. Verify:
 - All routers have correct OSPF neighbors
 - All PCs can ping each other
 - All PCs can ping the ISP (8.8.8.8)
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Bonus Challenges:

- Configure OSPF summarization at Area Border Routers (ABRs)
 - Set different OSPF costs on certain links
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Would you like me to:

- Give you a **recommended VLSM / IP plan** first?
- Provide a **step-by-step configuration guide**?
- Or start with just the blank table for you to design?

Let me know how you want to proceed!